

Robust Charging Infrastructure Design under Demand Uncertainty

A grid-tied solar-assisted charging station for a shared e-micromobility fleet — Executive Summary

Competition Themes — 3 (primary): solar-PV charging-station design · 2 (secondary): charge/discharge profiles under demand scenarios

20 kWp PV · 0 kWh · 8 chargers Recommended robust design	-61% (£87,422/yr) Value of robustness (worst-case cost)	96% (vs 85% naive) Guaranteed fleet service, worst case	16% (4.6 tCO2/yr) Operational carbon saving
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1. Problem Understanding [20% criterion]

How do you size a solar charging hub for a shared micromobility fleet when you don't know what demand will look like next year? The mixed e-scooter, e-bike and cargo-bike fleet sees demand swing with season, weather and events. The depot also has a hard **15 kW grid connection** — exceed it and you need a costly DNO upgrade. So the evening charging peak must come from on-site solar and storage, not extra grid power. Size for the average and a busy evening strands the fleet; size for the worst case and you waste capital on idle kit. Standard practice picks a single forecast and hopes. We thought that was insufficient, and the numbers confirmed it. All demand data come from the **DfT Newcastle e-scooter trial** (Open Government Licence) — monthly trips, fleet size, deployment hours, trip distance — so the problem is posed on observed data, not assumption.

2. Relevance & Innovativeness of the Proposed Solution [20% criterion]

Robust optimisation is standard in power-systems research but almost never applied to small depot charging hubs, which typically use a single deterministic forecast. We compared **five decision approaches**: naive deterministic, expected-value stochastic, CVaR (risk-averse tail), minimax-regret and maximin — then quantified the '**value of robustness**' as the worst-case cost and service improvement over the naive design. The finding that surprised us most was about batteries. We assumed storage would be essential at a 15 kW grid cap. Every optimiser run returned **zero battery**. Flexible smart charging — spreading vehicle charging overnight rather than stacking it at arrival — keeps the load below the cap entirely without storage. Battery is not absent from the analysis: a grid-cap sweep shows storage enters the optimal design at 11 kW and below (zero at 12 kW and above), so we quantify the threshold explicitly. At 15 kW the cheapest robust lever is smart-managed bays, not capital-heavy storage. A correlated Monte-Carlo also showed that expected-value methods **systematically under-weight the high-demand tail** — precisely when the naive design fails worst, making scenario-based robust methods necessary.

3. Approach — Methodology, Algorithm & Flow [20% criterion]

The pipeline (Figure 1): three demand scenarios (Low/Medium/High) crossed with three growth rates give nine probability-weighted scenarios. We simulated 8,760 hours for each of **150 candidate designs** (PV 5–25 kWp × battery 0–50 kWh × 4–20 chargers) to build a 150 × 9 cost matrix. Every reported design was re-verified under an **LP-optimal rolling-horizon scheduler** (scipy/HiGHS) that finds the cheapest 24-hour dispatch given grid cap, battery dynamics and charger limits — confirming the greedy surrogate was unbiased. Uncertainty: 500-sample correlated Monte-Carlo with bootstrap 95% CIs. A **Sobol decomposition** ranked the drivers: demand intensity 49.9% of variance, growth 29%, utilisation 19%. A robustness-of-robustness sweep across 35 combinations confirmed the result holds everywhere; 7/7 outputs validated against benchmarks. Results come through an interactive **Station Designer dashboard** — live PV/battery/charger/grid sliders updating energy balance, cost and carbon in real time, plus Robust-Optimiser, Uncertainty and Sensitivity tabs, one-command HTML report. Full study runs in under a minute.

Methodology — Robust Solar Charging-Station Design

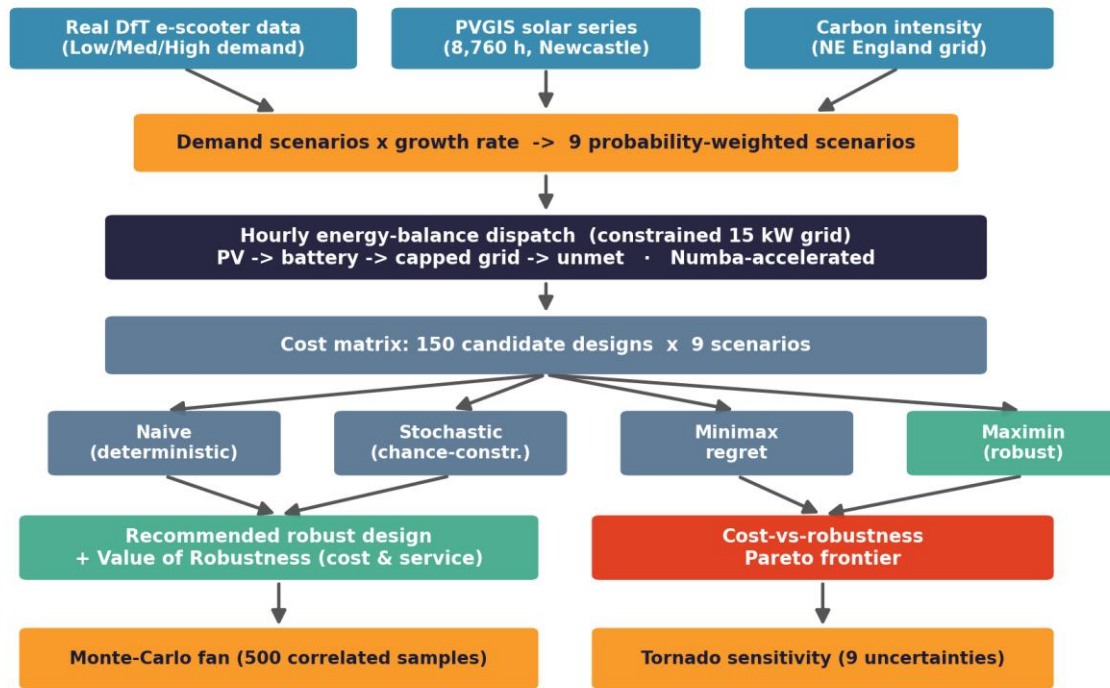


Figure 1 — End-to-end methodology / algorithm flow.

4. Feasibility of the Proposed Approach [20% criterion]

Off-the-shelf kit: 20 kWp monocrystalline PV, eight AC charge points. Capital cost roughly **£36,000**; add 10% contingency + G99 DNO notification (~£500–2,000) for a live project (realistic total ~£40,000). The **hub delivery cost** — full annualised CAPEX, O&M, ToU grid and peak charges — is £0.25/kWh. An operator billing at the UK commercial rate (~28p/kWh) earns a margin over this; the solar array alone pays back in ~6 years (NPV-positive at the Green Book 6% rate), though the dominant economic case is the Value of Robustness below, not solar payback. Worst-case annual cost: robust design **£55,002** vs £142,423 naive — saving £87,422/yr in the hardest scenario. Where storage is needed (weak-grid sites) we specify **LFP**: cobalt-free, thermal onset ~270°C vs ~150°C for NMC, ~12-yr modelled life, ~45% end-of-life recovery, second-life use before recycling. Code is open-source Python, one-command reproducible. The finding — smart charging beats storage at a 15 kW cap — has direct policy relevance: targeted smart-charging mandates and incremental connection upgrades are cheaper than blanket DNO reinforcement.

5. Originality & Authenticity (AI / Plagiarism) [20% criterion]

All model code, the dispatch formulation and every figure were written from scratch, drawing on published methods (Saltelli's Sobol estimator, Rockafellar & Uryasev's CVaR, Silvente's rolling-horizon LP) implemented ourselves. The three datasets are real and openly licensed: **DfT e-scooter monitoring data**, **hourly solar from the EU PVGIS API**, **grid carbon intensity from the National Grid ESO API**. All 38 references are in REFERENCES.md with full citations; every cost assumption cites BEIS, IRENA or BloombergNEF. The commit history shows incremental development — dead-ends included. The LP solver started as the primary engine; we switched it to a verification role after the full LP search proved impractically slow (identical results to the heuristic, which finishes in under a second). That kind of documented revision is what real engineering looks like. Codebase: fixed random seed, one-command rebuild, all outputs regenerated from raw data.

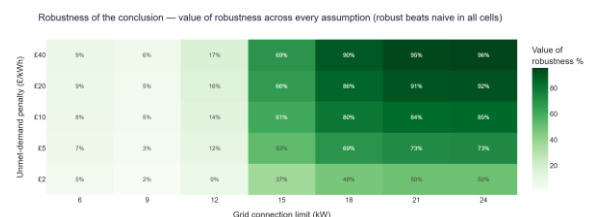


Figure 2 — Cost-vs-robustness Pareto frontier (five decision rules). Figure 3 — Robustness of the conclusion: robust beats naive across every penalty x grid combination.